Filling Level Estimation with Neural Networks

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ABSTRACT

Existing filling level estimation solutions are mostly designed for specifically defined industrial using scenario. For daily filling scenario, the problem is more challenging due to the variance of the problem space. Here we present a audio-based model based on neural networks which uses raw waveform sound to estimate the filling level. Aleatoric uncertainty is modeled as well to improve the performance of the model. The proposed model is shown to outperform the baseline machine learning approaches.

CCS CONCEPTS

• **Computing methodologies** → **Neural networks**; *Supervised learning by regression*; Kernel methods.

KEYWORDS

neural networks, filling level estimation

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1 INTRODUCTION

Filling level estimation (e.g., tank level monitoring) is vital for automatization of industries. There are different studies addressing this issue with magnetic field response sensor [36], IR imaging [38], or optical fibers [24]. Such solutions can be intensive in computation or intrusive to the container. There are also non-intrusive solutions based on audio signals. In one study, a Helmholtz resonator was used to determine the liquid volume inside the tank [33]. In another study, the resonant frequency of the tank was measured to estimate its filling level [11].

In the daily scenario, however, the problem is not always as well-defined as it is in the industrial cases. For example, estimating the filling level is useful for a coffee machine to avoid overfilling mugs with arbitrary sizes. The solution should be non-intrusive to the mugs. It will also be challenging for the existing audio-based solutions due to the lack of information about the containers. The dimension of the container and the properties of the flow, are physically related to how the sound is generated. Without such information, deriving an accurate estimation as it is done in previous audio-based works is intractable. Different from the traditional model-driven methods, data-driven methods such as machine learning will be a feasible way to capture the relationship between the acoustic information and the filling level with uncertainty.

Although there are uncertainties in the environment, human as perceivers can harness potential information from structured stimulation in the environment to better infer about it [14]. For example, the fundamental resonance frequency increases as a container fills with water, and it is found that human can estimate the filling level from the auditory information [4]. In a similar scenario, changes in the temperature of water leads to changes in its viscosity, which results in changes in the sonic properties when the water is poured into another container [12]. It is found that human can even perceive such difference from sound to infer the temperature of the pouring water [31].

In this project, we propose a neural network model to estimate the filling level of liquid from audio signals. We use multi-task learning strategy to estimate the uncertainty of each prediction to improve the performance of the model. As far as we could find, there was no available dataset and published research work on the same scenario before our implementation, we collected data on our own and compare the proposed model with traditional machine learning models.

2 RELATED WORKS

2.1 Conventional sound recognition approaches

Conventionally regression models and classification models like support vector machine (SVM) are applied to the extracted features to address sound recognition problems [6, 22, 23, 30]. Hand-crafted features such as log-mel feartures, Mel frequency cepstral coefficients (MFCCs) and linear prediction coefficients(LPCs) are widely used. *openSMILE* [8] is proposed as a paradigm for audio parameterization which generate audio parameters for each segment of recording, including MFCCs, LPCs, zero- and mean-crossing rate, energy, intensity /loudness, fundamental frequency and and etc. Then functionals like percentiles, mean, peaks and etc. are applied to each generated feature. With *openSMILE*, more than 6000 features can be generated from a clip of recordings depending on the configuration.

Feature selection is always important to improve efficiency and avoid overfitting [34]. Relief [19] and its extension ReliefF [20] were proposed to estimate the quality of attributes with awareness if contextual information by giving a high score to features whose value is consistent for instances with the same label and varies across instances with different labels. Their extension RreliefF [26] was proposed to addressing attribute estimation for regression problems. Previous research show the performance of parameterization

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by *openSMILE* and feature selection by RRliefF on human height regression form speech [9].

2.2 Sound recognition networks

Similar to conventional machine learning approaches, log-mel features are widely used as input of CNNs [1, 16, 25]. Log-mel filter banks are inspired by how human perceive audio signals [7] and it is argued that human physiological evidence may not be guaranteed the best filter banks for all types of problems [27, 28]. Raw waveform are proposed to be input of networks so that the networks can learn to generate features directly from the sound, and it is shown that distinguishable features can be learned with between-class learning strategy [29].

Pouring sound neural networks (PSNN) is proposed recently to estimate the poured liquid weight with multimodal convolutional networks (CNNs) to help pouring control of robots [35]. The mel-scaled spectrum of sound of 0.2 second interval is taken as the input of their audio-based network (PSNN-A), and the target is to estimate the poured weight by classification on classes of 0.2 oz increments. Our proposed model addresses the similar problem in a different way: we use raw waveform instead of mel-scaled spectrum as input and we regress the filling level instead of classify the poured weight, so that task-relevant features can be learned and the model can be generated to different containers. Additionally, the PSNN work make use of both video and audio input and integrate the information from both sides, which is fundamentally different from ours with only audio input. Besides, we care more about the filling level instead of just the weight and if it's going to overflow.

2.3 Uncertainty estimation

Uncertainty aware training has been shown to improve the performance for both classification tasks, regression tasks and reinforcement learning tasks [15, 18]. The advantage of uncertainty aware training is not just limited to improving the accuracy, and it can also tell when the prediction made by the neural network model is unreliable, which is a very critical part for some applications like autonomous driving. To actually model uncertainties with neural network, the neural network is viewed from a perspective of Bayesian probability theory, which is called as Bayesian deep learning. In this project, we model the aleatoric uncertainty by modifying the training objective of the neural network with a multi-task strategy, which is proposed by [18].

3 DATA COLLECTION

We collected all the audio tracks using the microphone of Google Pixel 2 unlocked smartphone with the application Voice Recorder, sampling at the rate of 44.1 KHz. To mitigate the influence of background noise, we selected a relatively quite room, ICCS 148, to perform all the experiments. So far, we collected a total of 46 episodes of water filling audio data. In each episode, we fill the container from complete empty to full, however, when to stop filling in each episode fully depends on personal judgement, and is inevitable to human error, which we will discuss more in the Limitations section.

During each episode, we try to maintain the flow rate of the water to be constant, but across different episodes, we fill up the containers with different flow rate to add more diversities to the dataset. We recorded the audio data with five different containers to encourage and evaluate the generalizability of our machine learning models, among which two of them are mugs, another two are made of stainless steel, and the other one is made of plastic. We split up the 46 collected episodes into training set, validation set and test set, and there are 30, 8 and 8 episodes in training, validation and test set respectively. The audio track with the Contingo bottle (top left one in Figure 1) is only contained in the test set not in the training set, which aims at evaluating the performance of generalization to difference containers of the machine learning model. In terms of water dispenser, we stick with the one in ICCS 148 because it produces distinguishable water filling sound consistently. We will discuss more regarding the water dispenser in Section **??**.

Figure 1: 5 different containers used to the experiments

4 DATA ANNOTATION

We assumed that the liquid flow was constant within each trial, so the filling level in percentage is a linear function of the time point t , i.e., $L(t) = \frac{t-t_s}{t_e-t_s}$, where the filling process starts at t_s and ends at t_e.

As in Figure 2, the filling sound is relatively loud in the recordings, so we segment the filling process with k-means cluster method based on the loudness. We apply short-time Fourier transform (STFT) on the original recordings with a window size of 0.1s and calculate the relative amplitude in terms of dB. To mitigate the effect of noise while preserve the change in the curve, we apply median filter (kernel size = 7) to the curves.

Note that the beginning filling sound is not always as loud as the following ones, especially for the recordings with the mugs, because there is not much water in the container yet to generate loud splashing sound. As it can be seen from Figure, 2, the background sound after filling is always louder than the sound before filling.

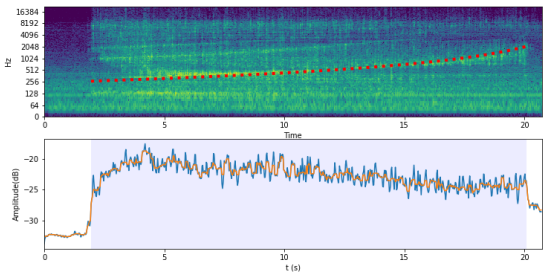

Figure 2: Top: Spectrum of a filling recording, with red dashed line marking the approximated Helmholtz resonant frequency. Bottom: The original (blue) and median-filtered (orange) amplitude of the recording, where the filling interval is shaded.

To avoid mis-classification of filling and background sound, we applied k-means on the first and second half of a recording respectively. We also use different weights to determine the threshold of cluster boundary since the variance of background sounds is smaller than the filling sounds. The thresholds are computed as a weighted average amplitude of two clusters, i.e $A_{threshold} = wA_b + (1-w)A_f$, where A_b is the cluster center of the background cluster and A_f is the filling one. We use 0.6 and 0.7 for weights of the first and second halves respectively. The second segmentation method works better for all the recordings.

With the filling percentage annotation we can also label the data in other metrics with the assumption that the shape of the body are approximated as cylinders. With S_i and l_i as the cross-section area and length of the body of container i and $l_p i$ as the neck length, the filling height is $h(t) = L(t)l_i$, the height left is $h_l(t) = (1 - L(t))l_i + l_{pi}$, and the filled volume $V(t) = L(t)S_i l_i$.

5 TRADITIONAL MACHINE LEARNING APPROACHES

There are a few seemingly reasonable options for the regression target of the model: the percentage of the volume has been filled, the height of the filling level, the weight/volume of the filled water, and the logarithm of the height left. The logarithm is a decent choice because theoretically the relationship between height and fundamental frequency is defined as:

$$f_{res} = \frac{c}{2\pi}\sqrt{\frac{S_p}{V_c l_p}} \approx \frac{c}{2\pi l_p}$$

where c is the speed of sound in gas, S_p is the cross section area of container neck, V_c is the cavity volume and l_p is the neck length [32][1]. For a bottle where the neck area is big enough, we approximate the cavity as an open ended cylinder, i.e., $V_c \approx S_p l_p$. The approximated resonant frequency curve is marked with red dashed line in Figure 2. Note that it overlaps with a formant but doesn't matches exactly, especially at the beginning of filling. Taking logarithm of both sides of the formula can result a linear relationship that $\log f_{res} \approx \log \frac{c}{2\pi} - \log l_p$. Therefore, we fit same machine learning models for those four different criteria. However, it turns out that the trained machine learning models achieve the best performance for the percentage of the volume has been filled; hence, in the rest of the project, the percentage will be the main target for the regression methods.

Since previous research shows correlation between water filling level and fundamental frequency of water filling sound, our initial attempt is augmenting the dataset with spectrum and cepstrum features, and applying feature selection algorithm on the dataset. We use a window size of 0.5 seconds to segment the data as we find that a too small window size will cause water filling sound hard to distinguish for human. Specifically, we decide to apply *OpenSMILE* [8] feature extraction toolbox, which is originally designed for human speech recognition task. We applied *OpenSMILE* to generate 6373 features in total. Feature selection is necessary because the high dimension data will cause severe overfitting, and models can perform poorly on test data. Without feature selection, the machine learning model predicts a constant value on the test data.

We applied RreliefF [26] feature selection algorithm to our data, which is a heuristic-based feature selection algorithm based on nearest instances of each data point. By applying this feature selection algorithm, the features with the highest scores are dominated by MFCC based features. The top 15 features are shown in Figure **??**.

We selected the top 120 features to test the performance. As Table 1 shows, the training and test loss for three methods measured as L1 loss, among the three regression methods applied, linear regression achieves a slightly better performance but still far from satisfaction. Due to the curse of dimensionality, Gaussian process and SVR cannot fit the dataset without feature selection. The label in the dataset has a uniform dataset; thus, a dummy predictor which simply predicts the expected percentage of the dataset (0.5) will achieve an error rate of 0.25. However, the best performance on the test set is 0.17 by linear regression which is just slightly above the dummy baseline.

Table 1: Training error and test error

Methods	Feature Selection	Training Error	Test Error
Linear Regression	Yes	0.168	0.175
	No	0.007	0.341
SVR	Yes	0.09	0.24
	No	-	-
Gaussian Process	Yes	0	0.483
	No	-	-

6 DEEP LEARNING BASED APPROACHES

Since traditional regression methods are insufficient to solve the problem, we decide to switch to more powerful deep learning approaches. In traditional machine learning, feature extraction and

[1]Technically, an effective neck length for an arbitrary Helmholtz resonator should be corrected as $l'_p = l_p + 0.6r + \frac{8}{3\pi}r$, where r is the radius of container neck [2, 5, 17].

Table 2: Top 15 features selected via RReliefF

Feature Name	Score
mfcc_sma[3]_rqmea	0.368
mfcc_sma[9]_percentile99.0	0.307
mfcc_sma[3]_quartile1	0.296
mfcc_sma[3]_quartile3	0.296
mfcc_sma[2]_percentile99.0	0.296
mfcc_sma[2]_quartile3	0.285
mfcc_sma[2]_amean	0.283
mfcc_sma[9]_peakMeanAbs	0.280
mfcc_sma[2]_quartile2	0.277
mfcc_sma[3]_amean	0.271
mfcc_sma[4]_amean	0.269
mfcc_sma[4]_quartile1	0.263
mfcc_sma[2]_peakMeanAbs	0.259
mfcc_sma[3]_peakMeanAbs	0.257
mfcc_sma[2]_rqmean	0.247

feature selection plays an important role in the success of the learning algorithm. Despite the fact that *OpenSMILE* adds a wide range of human-engineered features to the dataset, we still think that deep learning method can find a better feature representation for our dataset.

6.1 EnvNet

There is no existing neural network architecture designed specifically for the exact same problem that we face in our project; therefore, we slightly adapt existing neural network architecture to fit the objective of our project. EnvNet [28] is a end-to-end convolutional architecture designed to address the problem of environmental sound classification [25]. We think it is worthwhile to try EnvNet to regress to the filling level because water filling sound is more like a kind of environmental sound than human speech.

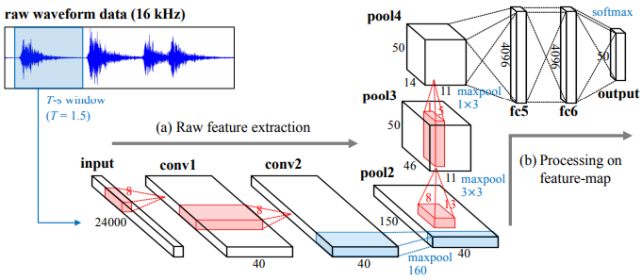

Figure 3: EnvNet

The original EnvNet can be viewed as a combination of two modules: feature extraction part and classification part. Instead of using popular MFCCs [37] or log-mel spectrum[35] as input, EnvNet directly feeds raw wave data into the neural network.

The first two layers make use of 1D convolutional filters to extract local features, followed by non-overlapping max pooling layers. Since we choose a window size of 0.5 seconds and the sampling rate is 44.1 KHz, the input size is a 22K long vector which is quite close to the 24K long input of the original EnvNet. Thus, we simply apply the same number of filters as the original paper. After the two 1D convolutional layers, the units in pool2 layer can be treated as a feature map. Suppose the feature map has a dimension of 150×40 as Figure 3., the 40 units in one row can treated as different features extracted from the same data while extracted features across the other dimension represent temporal information of the input data.

After the feature extraction part, the feature map can be processed similarly as a 2D image. Hence, similar to a classical CNNs for image classification [21], there are two layers of 2D convolutional filters with max pooling to extract local features from the feature map. Then, the extracted features are flattened and fed through two fully-connected layers.

After the fully-connected layers, the original last layer consists of 50 units represent the probability of each class for classification. To adapt the structure for regression problem, we just use one unit for the last layer. In addition, we add sigmoid activation function to it if we want the output to be percentage represented as decimal numbers between 0 and 1, which stabilizes the output and improves the accuracy on the validation set.

Since all the traditional machine learning methods failed due to overfitting and poor generalizability, adding both explicit and implicit regularizers is very important. Dropout is added to the last two fully connected layer with a probability of 0.5 to drop a neuron. In addition, we add weight decaying as another regularizer to mitigate overfitting problem. Following all the convolutional layers, there are batchnorm layers applied, which can be treated as a kind of implicit regulizer. We use Adam as optimizer with a learning rate of 10^{-3} for all the training we do in this project. We also record the validation loss along with the training process to implement early stopping, which we expect to stop training when there is a obvious rise in the valition loss. In fact, we train the adapted EnvNet architecture for 2000 epochs, but we do not find such a corner point where validation error suddenly starts rising.

6.2 Modelling Uncertainty with Neural Network

In previous research, it has been shown that modelling uncertainties with neural networks has a few benefits including avoiding dangerous decision made by the neural network and improving the prediction accuracy in general [18].

There are actually several different types of uncertainty, among which the most important two are aleatoric and epistemic uncertainty [3]. Epistemic uncertainty is also commonly named as model uncertainty, which is usually crucial to model for a small and sparse dataset. Additionally, aleatoric uncertainty represents uncertainty with respect to information which our data cannot explain. Equivalently, it is a representative for factors that differ every time we run the exact same experiment.

In our case, modeling aleatoric uncertainty is certainly beneficial. As we do not have well-designed hardware and accurate sensors to ensure the measuring process perfectly consistent across different trials, it is very likely that similar audio segment will be labeled

quite differently. There are a few factors contributing to aleatoric uncertainty. For example, the annotation of the audio data is highly dependent on the assumption of constant flow rate during each episode. Across different trials, when exactly we stop filling is purely based on human judgement. Thus, although we label the beginning of each trial as 0 percentage and the end as 100 percentage, those labels yet involve instinct human errors. Besides the uncertainty caused by annotation, some other factors like the place we place the bottle and the orientation and position of the mobile phone also vary across different trials, which are not observed explicitly by the data.

To actually model aleatoric uncertainty with neural network, the output of the neural network is no longer a single numerical value representing the filling level. Instead, the output will be actually modeled like a normal distribution [18]. Specifically, we need to add another unit to the output layer such that the two units can reflect the mean and log-variance of the distribution. In addition to the structural change, the loss function needs to be modified correspondingly.

$$Loss = \frac{\|y - \hat{y}\|_2}{\sigma^2} + \frac{1}{2}\log\sigma^2$$

The first term in the loss function is the commonly used mean squared loss scaled by the variance, and the second term as a regularizer prevents the uncertainty term growing infinitely large. Intuitively, if the predicted value $\hat{y}$ is far away form the label y, the variance term will be encouraged to be large to scale down the large mean squared loss.

To verify that modelling aleatoric uncertainty is beneficial for our project, we plot the validation loss for both cases when aleatoric uncertainty is modelled or not during the training. For the interest of time, we train both methods for 2000 epochs, which we think is sufficient for the verification purpose. In Figure 4, the X and Y axes reflect the training epochs and the validation loss using L1 error $\|y - \hat{y}\|_1$ respectively. The orange dashed line belongs to the case without modelling aleatoric uncertainty while the blue line represents the validation loss with uncertainty modelled. When aleatoric uncertainty is modelled with the EnvNet, the neural network produces more stable outputs and also obtains lower validation loss. Without aleatoric uncertainty modelled. the validation loss fluctuates greatly. Based on the plot here, we reckon that modelling aleatoric uncertainty can improve the performance of the network and also generalize well on the test set.

7 DOMAIN REGULARIZER

Besides modelling aleatoric uncertainty with neural network, we also add a discriminator part to the EnvNet architecture, which aims at learning domain-invariant features from the feature extraction stage. As we fully aware that our small training dataset may not cover the true distribution of real world data, we hope to improve the robustness of our model to distribution shift. By adopting the idea of domain-adversarial training [10] (see Figure 5), the objective of the training is decomposed into two parts: (1) minimizing the regression loss, and (2) maximizing the loss of the discriminator. The second part of the objective is to fool the discriminator such that the discriminator cannot distinguish the label from the extracted features. Therefore, such a fooled discriminator can be treated as a

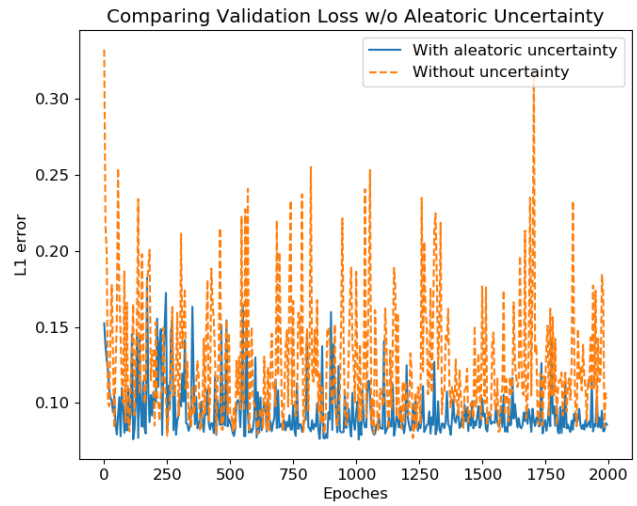

Figure 4: Training with and without aleatoric uncertainty

domain regularizer, which encourage the feature extraction module to learn domain-invariant features.

In our case, the discriminator is designed to unsuccessfully distinguish the type of container so that the model could perform well for different containers. This can be easily extended to other kinds of domain shift, including different water dispensers. The value of β is a hyperparameter that needs to be tuned manually, and we find that setting it to 10^{-2} works well in our case.

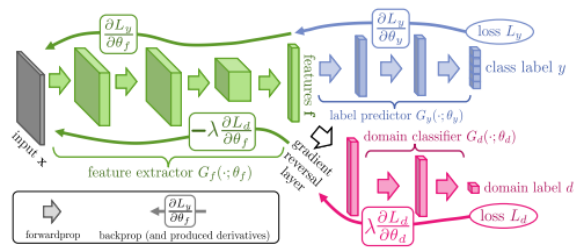

Figure 5: Architecture of domain-adversarial neural networks

8 RESULTS

In Figure 6, we plot the performance of model on the test data when the EnvNet architecture is modelled with aleatoric uncertainty. In Table 3, we include the accuracy of models in mean absolute error (MAE) with two different ways of annotation that we tried in this project. The simple k-means approach applied clustering on the whole filtered loudness curve which introduces more error in annotation than applying cluster on the two halves separately. Note that even simple k-means introduce errors in annotation, modeling aleatoric lead to better performance of EnvNet.

We choose four episodes to show in the report since the performance of episodes with the sample container is quite similar, but

Table 3: Accuracy of different experiments

Model	Annotation	Test Error
Linear Regression	two-halves k-means	0.175
EnvNet	simple k-means	0.1279
EnvNet w/ uncertainty	simple k-means	0.0964
EnvNet w/ uncertainty	two-halves k-means	**0.0942**
EnvNet w/ uncertainty & discriminator	two-halves k-means	0.0996

different containers do affect the accuracy of the prediction. The top two plots are based on experiments of the Thermos bottle(the right one of the second row in Figure 1) and UA bottle(the middle on of the first row in Figure 1). In those two cases, the prediction follows the label closely, which meets our expectation. Similarly, the model delivers decent performance for the data collected with the Contigo bottle (top left one in Figure 1). As we mentioned in Data Collection section, the data collected with Contigo bottle is not included in the training set. Thus, this plot shows the generalizability to different containers of the EnvNet model. However, we are certainly aware that our model still fall short of generalizing to a lot of complicated setups.

The last plot(bottom right) in Figure 6 is with the mug in the first row of Figure 1. In this case, the performance is not as good as we expect since there are few predictions that are far away from the label. The data collected with mugs seems to be harder to predict than other containers because the mug is usually smaller and faster to fill up. Therefore it is prone to the inconsistency of the annotation, and the two-halves k-means clustering improve the results of mugs but cannot completely solve the problem. Nevertheless, the prediction is not trivial because most prediction still follows the trend of the label, whose value increases as more portion of the container is filled.

In addition, adding domain regularizer to the neural network architecture does not reveal huge difference. Domain regularizer does not show its advantage because most test data and training data are collected using the same setup. Thus, most of the training data and the test data originates from the same distribution, in which case domain regularizer does not boost the performance of the model since it is designed for overcoming problems involving large distribution shift. However, we still think this approach should be adopted because our dataset only covers a tiny subset of the water filling sound in real world. There will always be distribution shift in almost all scenarios with different setups. Thus, we expect training with this domain regularizer can effectively alleviate the pain of distribution shift.

9 LIMITATIONS AND FUTURE WORK

9.1 Human error in annotation

The filling level measurement is highly dependent on the constant flow rate assumption, which is not always true in the data collection process. A more reliable and accurate data collection setup can lead to a dataset with less noise and the data points are more consistent with each other. To improve the data collection process, we could setup a electronic scale and a data acquisition system to measure the weight of the water filled in the containers [35]. The weight of the filled fluid can be easily transformed into the quantity that we want to regress to, including percentage of filled water, height of the filling level and etc. if the dimension of the container is known or measured.

Besides measuring filling level, the timing to stop fill the container is also hard to decide for human. Only having audio data afterwards makes it impossible for us to decide when exactly the container is full. One way to address this for our project is recording video data directly, using the visual data to determine the cut-off points, and then extracting the audio track as the input to the neural network.

9.2 Distribution shift

As we mentioned in the data collection section, our data is collected with the same water dispenser and 5 different containers made of different materials. Therefore the dataset is relatively homogeneous. There are a countless number of combinations of different water dispensers and containers for the water filling process. Therefore, the distribution of our dataset may not reflect the true distribution of audio during the water filling process. Although we adopt some regularization techniques like dropout, weight decay and learning a discriminator as a domain regularizer. However, due to the time constraint, we do not have time to collect data in various environments and test how well our model can generalize to different environments and setups.

9.3 Noise Reduction

All the data we collected is in a quiet environment when water filling sound can be clearly identified by human. However, noise is almost everywhere in real life, which can deteriorate the performance of the model drastically. We could train a denoising autoencoder to partially solve the problem of noise. We can synthesize the training data by mix the collected water filling acoustic data with some audio dataset like Google AudioSet [13]. Ideally, the denoising autoencoder can learn a mapping between the noisy audio and the pure water filling sound.

10 CONCLUSION

In this project, we try to predict the filling level when filling the container with water. Our initial attempt by using traditional regression methods is not successful since the test error is not much better than the dummy baseline which predicts the average of the dataset. We then slightly adapt the EnvNet to regression problem, which is originally designed for environmental sound classification problem. Also, we model the aleatoric uncertainty with the neural network architecture which improves the accuracy on the test set. In addition, we implemented a discriminator to encourage to learn a domain-invariant feature space such that the model is expected to perform better with data collected with different containers and water dispensers.

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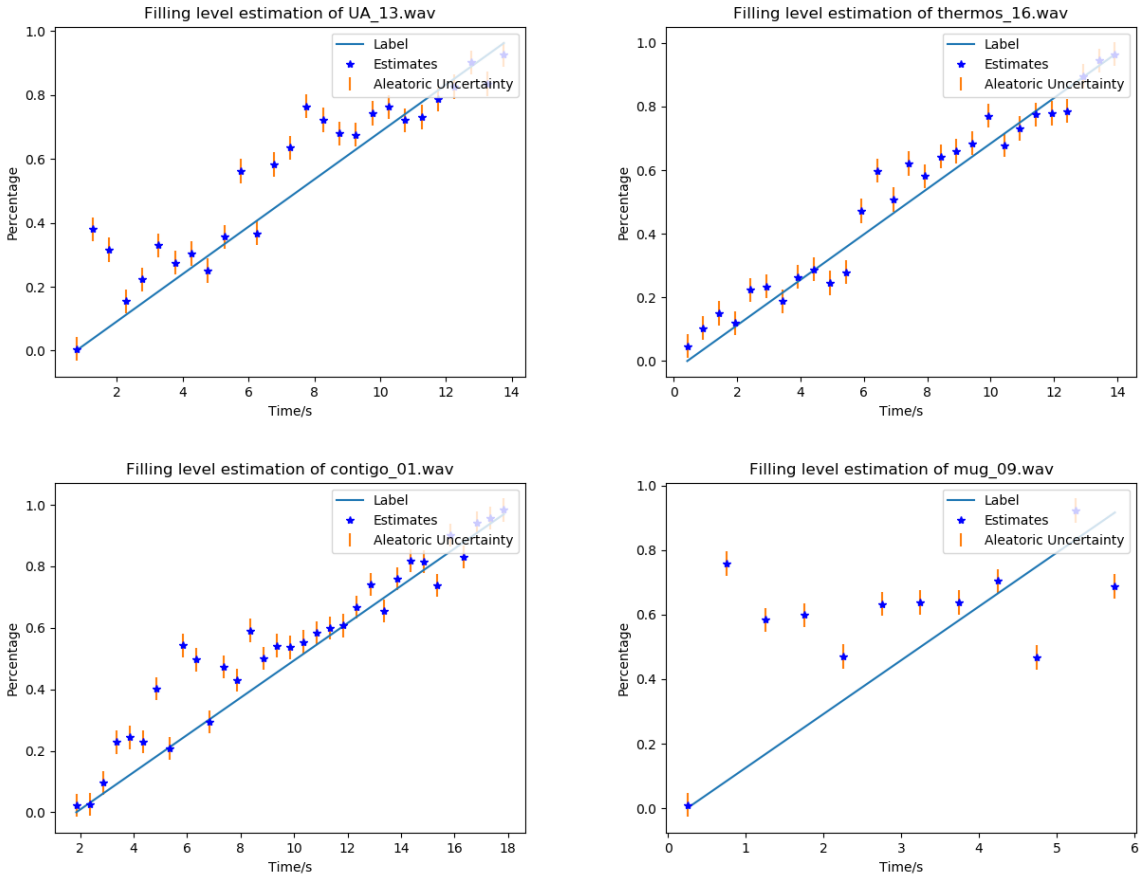

Figure 6: Test cases performance

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